

High-Dimensional Epidemic Dynamics, Stability, Oscillations, and Probabilistic Outcomes

Pramithas Upreti

Department of Mathematics

Abstract

Compartmental epidemic models have long served as foundational tools in mathematical epidemiology [1, 11, 10]. Classical models such as the SIR framework provide clean insight into threshold behavior and disease transmission [1, 19], but they frequently fail to capture mechanisms observed in real outbreaks: latent infection, hospitalization, environmental persistence, and delayed nonlinear feedback [17, 16, 22]. In this paper, we study an extended SEIHRV model consisting of six compartments—susceptible (S), exposed (E), infectious (I), hospitalized (H), recovered (R), and environmental viral load (V)—and rigorously compare its mathematical structure with the classical SIR system.

We derive the basic reproduction number $\mathcal{R}_0$ via the next-generation matrix method of van den Driessche and Watmough [3], prove global asymptotic stability of the disease-free equilibrium using a Lyapunov function [5], and characterize the endemic equilibrium through the center manifold analysis [4] near the forward transcritical bifurcation at $\mathcal{R}_0 = 1$. We then investigate how higher dimensionality introduces complex eigenvalues and the possibility of Hopf bifurcation and sustained limit-cycle oscillations [12, 15], phenomena impossible in the standard SIR system [7]. Finally, we develop a probabilistic framework in which the model parameters are treated as random variables, establish that identical initial conditions can produce qualitatively distinct epidemic trajectories, and use normalized forward sensitivity (elasticity) indices [6] to rank the relative influence of each parameter on $\mathcal{R}_0$.

The central theme is that, as the models become structurally more realistic, their evolution becomes sensitive, high-dimensional dynamics, and their behavior requires analysis using stability theory, bifurcation analysis, and probabilistic reasoning.

Contents

1	Introduction	3
2	The Classical SIR Model	4
2.1	Model Formulation	4
2.2	Basic Reproduction Number and Threshold Behavior	4
2.3	Bendixson–Dulac Criterion and Absence of Limit Cycles in SIR	5
2.4	Limitations	6
3	The SEIHRV Model	6
3.1	Compartments	6
3.2	Parameters	6
3.3	Compartment Diagram	6
3.4	Governing Equations	8
4	Biological Feasibility and Invariant Region	8
5	Equilibria	10
5.1	Disease-Free Equilibrium	10
5.2	Endemic Equilibrium	11
6	Reproduction Number via Next-Generation Matrix	11
6.1	Method	11
6.2	Computing V^{-1}	12
6.3	Next-Generation Matrix and $\mathcal{R}_0$	12
6.4	Epidemiological Decomposition	13
7	Local Stability Analysis	13
8	Global Stability of the Disease-Free Equilibrium	14
9	Center-Manifold Analysis and the Transcritical Bifurcation	15
9.1	Setup	15
9.2	Bifurcation Coefficients	15
10	Oscillatory Dynamics and Hopf Bifurcation	16
10.1	Mechanism for Oscillations	16
10.2	Hopf Bifurcation Condition	16
11	Sensitivity and Elasticity Analysis	18
11.1	Normalized Forward Sensitivity Index	18
11.2	Analytical Results	18

12 Probabilistic Framework and Parameter Uncertainty	18
12.1 Stochastic Parameter Model	19
12.2 Expected Trajectory and Variance	19
12.3 Distribution of $\mathcal{R}_0$	20
12.4 Delta Method Approximation	21
13 Numerical Simulations	21
13.1 Parameter Choices	21
13.2 Three Qualitative Regimes	21
13.3 Numerical Investigation of Hopf Bifurcation	23
14 Toward Global Stability of the Endemic Equilibrium	24
14.1 Numerical Evidence	25
14.2 Path to a Proof via Compound Matrices	26
14.3 Remarks	26
15 Comparison: SIR versus SEIHRV	27
16 Real-World Interpretation	27
17 Limitations	29
18 Conclusion	30
A Proof Details: Computation of V^{-1}	32
B Routh–Hurwitz Conditions for the Jacobian at $\mathcal{E}^*$	32
C Explicit Lyapunov Coefficient Computation	33
D Figure Attribution	33

1 Introduction

The quantitative modeling of infectious disease epidemics to describe, predict, and control the spread of an infectious disease constitutes mathematical epidemiology. The field’s recent roots go back to the paper by Kermack and McKendrick [1] that introduced the SIR model and introduced the notion of a threshold population size, which an epidemic cannot maintain if it is below.

The SIR model is simple and analytically tractable. It describes the epidemic growth, saturation, and decay in a three-dimensional autonomous system of ordinary differential equations (ODEs). It has the property of having only one parameter $\mathcal{R}_0 = \beta S_0/\gamma$ that is clearly interpretable, which will allow it to decide whether a disease will invade or die out. Despite this power, the SIR framework is based on assumptions that are commonly broken in the real world:

- *No latent period.* Individuals are assumed to become infectious immediately upon exposure, whereas most pathogens have an incubation period during which an infected individual is not yet contagious.
- *Homogeneous mixing.* Every susceptible individual contacts every infectious individual at the same rate, ignoring spatial structure and social heterogeneity.
- *No environmental reservoir.* Pathogens that persist outside the host—on surfaces, in aerosols, in water—are not modeled.
- *No disease severity.* All infected individuals progress through a single “infected” class; hospitalization, intensive care, and differential mortality are ignored.
- *Single epidemic wave.* Under standard assumptions, the SIR system has no mechanism for sustained oscillations or recurrent epidemic waves.

These deficiencies motivated extensions such as the SEIR model (adding an exposed class), the SIS model (allowing reinfection), and models with age structure, spatial heterogeneity, and stochastic noise [17, 19]. The SEIHRV model studied here incorporates several of these extensions simultaneously: an exposed class E for the latent period, a hospitalized class H for severe cases, and an environmental compartment V for indirect transmission via contaminated environments. Importantly, the compartment V is not a human population, but rather, it monitors copies of the pathogen (such as viral particles per volume) in a common abiotic environment, where individuals cannot be transferred to or from V in the demographic sense. These compartments make a 3-dimensional ODE system become a 6-dimensional nonlinear dynamical system. This dimensional change is not superficial, and it has many significant consequences for stability, bifurcation, and periodic orbits. The main contributions of this paper are as follows.

- We derive the SEIHRV system, check biological feasibility, and find the equilibria of the system.
- $\mathcal{R}_0$ is calculated using the next-generation matrix approach [2, 3] and broken up into ecologically meaningful transmission components.

- We prove local and global asymptotic stability of the disease-free equilibrium when $\mathcal{R}_0 < 1$, and describe the endemic equilibrium by the center manifold reduction when $\mathcal{R}_0 > 1$.
- We study the oscillatory nature, Hopf bifurcation conditions, and the occurrence of limit cycles.
- We present a probabilistic model with random parameters, discuss the resulting distribution of epidemic trajectories, and calculate normalized sensitivity (elasticity) indices for $\mathcal{R}_0$.

Throughout, we emphasize rigorous statements of hypotheses and proofs, building on standard references in dynamical systems [7, 8], mathematical biology [9, 10, 11], and bifurcation theory [12].

2 The Classical SIR Model

2.1 Model Formulation

Let N denote a closed, homogeneously mixing population of constant size. The SIR model partitions the population into three epidemiological classes:

$S(t)$: Susceptible, $I(t)$: Infectious, $R(t)$: Recovered/Removed.

The governing system, due originally to Kermack and McKendrick [1] and presented in modern compartmental form throughout [10, 11, 9], is

$$\begin{cases} \frac{dS}{dt} = -\beta SI, \\ \frac{dI}{dt} = \beta SI - \gamma I, \\ \frac{dR}{dt} = \gamma I, \end{cases} \quad (1)$$

with $\beta > 0$ (transmission rate) and $\gamma > 0$ (recovery rate). Since $\frac{d}{dt}(S + I + R) = 0$, the total population $N = S + I + R$ is conserved.

2.2 Basic Reproduction Number and Threshold Behavior

Define the basic reproduction number

$$\mathcal{R}_0^{\text{SIR}} = \frac{\beta S_0}{\gamma}, \quad (2)$$

where $S_0 = S(0)$ is the initial susceptible population. The quantity $\mathcal{R}_0$ represents the expected number of secondary infections produced by a single infectious individual introduced into a fully susceptible population. It governs epidemic behavior through the following classical result.

Theorem 2.1 (Threshold Theorem, [1]). *Let $(S(t), I(t), R(t))$ be a solution to (1) with $S(0) > 0$, $I(0) > 0$.*

(i) *If $\mathcal{R}_0 \leq 1$, then $I(t)$ is non-increasing and $I(t) \rightarrow 0$ as $t \rightarrow \infty$.*

(ii) *If $\mathcal{R}_0 > 1$, then $I(t)$ first increases to a maximum $I_{\max} = I(0) + S(0) - \frac{\gamma}{\beta}(1 + \ln(\beta S(0)/\gamma))$ and then decreases to zero.*

In both cases, $I(\infty) = 0$ and $S(\infty) > 0$.

Proof sketch. From (1), $\frac{dI}{dt} = I(\beta S - \gamma)$. Since S is strictly decreasing whenever $I > 0$, I is increasing if and only if $S > \gamma/\beta$. When $\mathcal{R}_0 \leq 1$, we have $S_0 \leq \gamma/\beta$, so $\dot{S}(0) \leq 0$ and S never rises; hence I never increases. Dividing the $\dot{S}$ equation by the $\dot{R}$ equation gives $\frac{dS}{dR} = -(\beta/\gamma)S$, yielding $S(t) = S_0 e^{-\beta R(t)/\gamma} > 0$ for all finite t ; therefore $I(\infty) = N - S(\infty) - R(\infty) = 0$. $\square$

Remark 2.2. A key geometric insight, classical in the dynamical systems literature [7, 8], is that SIR trajectories in the (S, I) phase plane are strictly monotone in S and exhibit no closed orbits. This follows immediately from Bendixson's criterion [7, Thm. 7.10]: the divergence of the SIR vector field is $\frac{\partial}{\partial S}(-\beta SI) + \frac{\partial}{\partial I}(\beta SI - \gamma I) = -\beta I - \gamma < 0$ for $I > 0$, so no closed trajectory can exist in the positive half-plane. We restate this result here because the parallel claim for the SEIHRV model (Section 10) will reach the *opposite* conclusion: limit cycles are possible.

2.3 Bendixson–Dulac Criterion and Absence of Limit Cycles in SIR

Theorem 2.3 (No periodic orbits in SIR). *System (1) has no periodic orbits in $\{S > 0, I > 0\}$.*

Proof. Let $\mathbf{f}(S, I) = (-\beta SI, \beta SI - \gamma I)$ be the SIR vector field restricted to the (S, I) plane. Its divergence is

$$\nabla \cdot \mathbf{f} = \frac{\partial(-\beta SI)}{\partial S} + \frac{\partial(\beta SI - \gamma I)}{\partial I} = -\beta I + \beta S - \gamma.$$

Since $\dot{S} = -\beta SI < 0$ whenever $I > 0$, S is strictly decreasing; but more directly, in the region $\{S > 0, I > 0\}$ we have $-\beta I < 0$ and $\beta S - \gamma$ may be positive or negative. A cleaner computation uses the Dulac function $h(S, I) = 1/I > 0$: then

$$\nabla \cdot (h\mathbf{f}) = \frac{\partial}{\partial S} \left(\frac{-\beta SI}{I} \right) + \frac{\partial}{\partial I} \left(\frac{\beta SI - \gamma I}{I} \right) = \frac{\partial(-\beta S)}{\partial S} + \frac{\partial(\beta S - \gamma)}{\partial I} = -\beta < 0.$$

By the Dulac–Bendixson criterion [7, Thm. 7.10], no closed trajectory exists in $\{S > 0, I > 0\}$. $\square$

This theorem is in sharp contrast with Proposition 10.2: that higher dimensionality is a necessary condition for sustained epidemic oscillations.

2.4 Limitations

The beauty of the SIR model in maths comes at the expense of biological reality. Its inability to represent latent periods, environmental transmission, hospitalization, or oscillatory dynamics motivate the construction of richer compartmental systems.

3 The SEIHRV Model

3.1 Compartments

We extend the SIR framework to six compartments:

$$S(t), E(t), I(t), H(t), R(t), V(t),$$

representing susceptible individuals, exposed (latent) individuals, actively infectious individuals, hospitalized patients, recovered individuals, and the environmental viral load (e.g., viral copies per unit volume in a shared environment), respectively. The population compartments satisfy $N(t) = S + E + I + H + R$.

3.2 Parameters

Before presenting the transfer diagram and governing equations, we list and briefly describe the parameters used throughout the model. Each parameter has a clear biological meaning: the β_i are per-capita transmission rates for each infection pathway, σ is the reciprocal of the average incubation period, γ and ρ are recovery rates from the infectious and hospitalized classes, δ is the rate at which infectious individuals are hospitalized, α is the rate at which infectious individuals shed pathogen into the environment, η is the environmental decay rate of the pathogen, μ is the per-capita natural mortality rate, and Λ is the *recruitment rate*—the rate (in individuals per unit time) at which new susceptibles enter the population through births or net immigration. Note that Λ is a total flow rate, not a per-capita rate; it is exactly balanced at demographic equilibrium by per-capita mortality μN , giving the disease-free population size $N^* = \Lambda/\mu$. The full list and the biological domain of each parameter is given in Table 1.

3.3 Compartment Diagram

The transfer diagram (Figure 1) shows how individuals move between the six compartments. Each arrow is labeled with the corresponding transition rate from Table 1. Susceptible individuals become exposed via three distinct routes (direct contact with I , presymptomatic contact with E , and environmental exposure to V); exposed individuals progress to infectious; infectious individuals either recover, are hospitalized, or die; hospitalized individuals recover (after a typically longer stay); and infectious individuals shed the pathogen into the environment, where it decays at a rate η .

Table 1: SEIHRV model parameters and biological interpretation.

Parameter	Description	Domain
Λ	Recruitment rate into S (births/immigration, per unit time)	$\mathbb{R}_{>0}$
β_1	Transmission rate: $S-I$ contact	$\mathbb{R}_{>0}$
β_2	Transmission rate: $S-E$ contact	$\mathbb{R}_{\geq 0}$
β_3	Environmental transmission rate	$\mathbb{R}_{\geq 0}$
σ	Progression rate $E \rightarrow I$ (incubation $^{-1}$)	$\mathbb{R}_{>0}$
γ	Recovery rate $I \rightarrow R$	$\mathbb{R}_{>0}$
δ	Hospitalization rate from I	$\mathbb{R}_{\geq 0}$
ρ	Recovery rate $H \rightarrow R$	$\mathbb{R}_{>0}$
α	Viral shedding rate into V	$\mathbb{R}_{\geq 0}$
η	Environmental decay rate of V	$\mathbb{R}_{>0}$
μ	Natural mortality rate	$\mathbb{R}_{>0}$

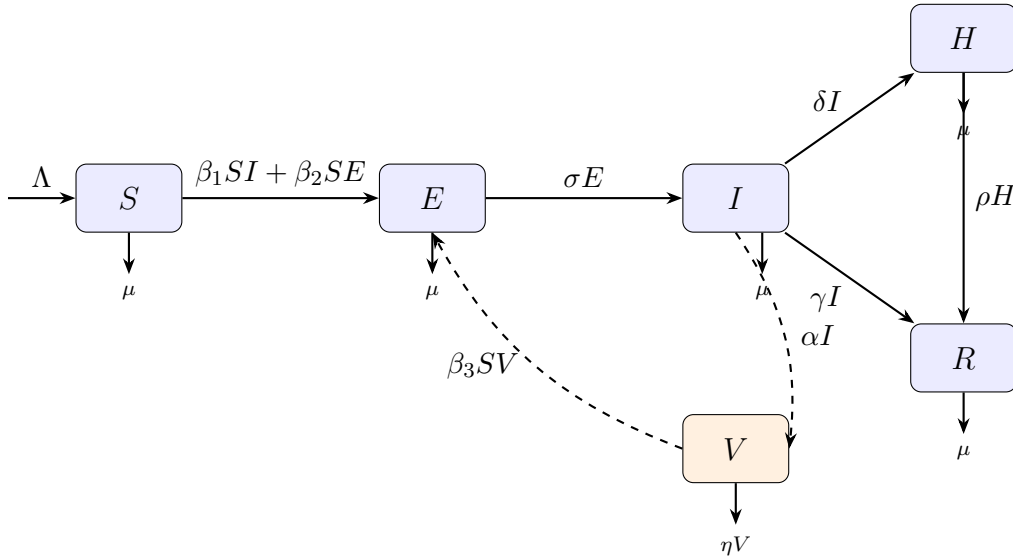


Figure 1: Transfer diagram for the SEIHRV model, drawn in TikZ [18] following the standard compartmental-diagram convention of the mathematical epidemiology literature [10, 11, 3]. Solid arrows represent transitions between human compartments (S, E, I, H, R); dashed arrows show environmental pathways involving the viral-load compartment V . **Note:** V is *not* a human compartment; it represents pathogen copies in the shared environment (e.g., fomites, aerosols). Each human compartment also experiences natural mortality at a rate μ (downward arrows, omitted from diagram for clarity). All parameters are defined in Table 1. Figure Attribution in Appendix D.

3.4 Governing Equations

The SEIHRV model is:

$$\left\{ \begin{array}{l} \frac{dS}{dt} = \Lambda - (\beta_1 I + \beta_2 E + \beta_3 V)S - \mu S, \\ \frac{dE}{dt} = (\beta_1 I + \beta_2 E + \beta_3 V)S - (\sigma + \mu)E, \\ \frac{dI}{dt} = \sigma E - (\gamma + \delta + \mu)I, \\ \frac{dH}{dt} = \delta I - (\rho + \mu)H, \\ \frac{dR}{dt} = \gamma I + \rho H - \mu R, \\ \frac{dV}{dt} = \alpha I - \eta V. \end{array} \right. \quad (3)$$

The force of infection acting on S is $\lambda(t) = \beta_1 I + \beta_2 E + \beta_3 V$, which couples three distinct transmission pathways into a single nonlinear term.

4 Biological Feasibility and Invariant Region

Before undertaking any stability or bifurcation analysis, we must verify that the model is *biologically well-posed*: all state variables must remain nonnegative (populations cannot be negative) and bounded (a finite population cannot grow without limit). This ensures that there is a physical interpretation for all the trajectories generated by the ODE system, and that the analysis we develop in the following sections is not vacuous. In a formal sense, we look for a region $\Omega \subset \mathbb{R}_{\geq 0}^6$ that is compact, and whose trajectories are uniformly forward-invariant.

Definition 4.1. The *biologically feasible region* is

$$\Omega = \left\{ (S, E, I, H, R, V) \in \mathbb{R}_{\geq 0}^6 : S + E + I + H + R \leq \frac{\Lambda}{\mu}, V \geq 0 \right\}.$$

The upper bound Λ/μ is the steady-state population size in the absence of disease-induced mortality: recruitment Λ is exactly balanced by natural mortality μN . The constraint $V \geq 0$ simply requires that the environmental viral load be nonnegative.

Theorem 4.2 (Positive Invariance). *The region Ω is positively invariant for system (3): every solution with initial data in Ω remains in Ω for all $t \geq 0$. In particular, solutions exist globally and are nonnegative and bounded.*

Proof. Let $N(t) = S + E + I + H + R$. Summing the first five equations of (3):

$$\frac{dN}{dt} = \Lambda - \mu(S + E + I + H + R) - (\gamma + \delta)I - \rho H \leq \Lambda - \mu N,$$

since $(\gamma + \delta)I + \rho H \geq 0$. This is a linear differential inequality with solution bounded by the Grönwall inequality [14, 7]:

$$N(t) \leq \frac{\Lambda}{\mu} + \left(N(0) - \frac{\Lambda}{\mu} \right) e^{-\mu t} \xrightarrow{t \rightarrow \infty} \frac{\Lambda}{\mu}.$$

Hence if $N(0) \leq \Lambda/\mu$, then $N(t) \leq \Lambda/\mu$ for all $t \geq 0$.

Nonnegativity of each state variable follows from the standard invariance argument for nonnegative systems [10, 19]: on each boundary face of $\mathbb{R}_{\geq 0}^6$, the vector field of (3) points inward or is tangent. Explicitly: $\dot{S}|_{S=0} = \Lambda > 0$; $\dot{E}|_{E=0} = \lambda S \geq 0$; $\dot{I}|_{I=0} = \sigma E \geq 0$; $\dot{H}|_{H=0} = \delta I \geq 0$; $\dot{R}|_{R=0} = \gamma I + \rho H \geq 0$; and $\dot{V}|_{V=0} = \alpha I \geq 0$. Nonnegativity of V also follows directly from $\dot{V} \geq -\eta V$ and $V(0) \geq 0$, giving $V(t) \geq V(0)e^{-\eta t} \geq 0$. $\square$

Corollary 4.3. *System (3) is well-posed on Ω : solutions exist globally and remain nonnegative and bounded.*

Proposition 4.4 (Uniform bound on environmental viral load). *Let $(S(t), E(t), I(t), H(t), R(t), V(t))$ be a solution of system (3) with initial condition in the feasible region Ω . Then the environmental viral load satisfies*

$$0 \leq V(t) \leq V(0)e^{-\eta t} + \frac{\alpha\Lambda}{\mu\eta}(1 - e^{-\eta t}) \quad \text{for all } t \geq 0.$$

In particular,

$$\limsup_{t \rightarrow \infty} V(t) \leq \frac{\alpha\Lambda}{\mu\eta}.$$

Thus $V(t)$ remains uniformly bounded by a quantity determined by viral shedding, population size, and environmental decay.

Proof. From the environmental equation in (3),

$$\frac{dV}{dt} = \alpha I - \eta V.$$

Since the solution remains in Ω by Theorem 4.2, we have

$$0 \leq I(t) \leq N(t) \leq \frac{\Lambda}{\mu}.$$

Therefore,

$$\frac{dV}{dt} \leq \frac{\alpha\Lambda}{\mu} - \eta V.$$

Equivalently,

$$\frac{dV}{dt} + \eta V \leq \frac{\alpha\Lambda}{\mu}.$$

Multiplying by the integrating factor $e^{\eta t}$ gives

$$\frac{d}{dt} (e^{\eta t} V(t)) \leq \frac{\alpha\Lambda}{\mu} e^{\eta t}.$$

Integrating from 0 to t ,

$$e^{\eta t} V(t) - V(0) \leq \frac{\alpha \Lambda}{\mu} \int_0^t e^{\eta s} ds.$$

Since

$$\int_0^t e^{\eta s} ds = \frac{e^{\eta t} - 1}{\eta},$$

we obtain

$$e^{\eta t} V(t) \leq V(0) + \frac{\alpha \Lambda}{\mu \eta} (e^{\eta t} - 1).$$

Dividing by $e^{\eta t}$ yields

$$V(t) \leq V(0)e^{-\eta t} + \frac{\alpha \Lambda}{\mu \eta} (1 - e^{-\eta t}).$$

The nonnegativity $V(t) \geq 0$ follows from the positive invariance argument in Theorem 4.2. Taking the limit superior as $t \rightarrow \infty$ gives

$$\limsup_{t \rightarrow \infty} V(t) \leq \frac{\alpha \Lambda}{\mu \eta}.$$

□

Remark 4.5. This bound has a direct biological interpretation. The environmental viral load cannot grow without limit because viral shedding is bounded above by the maximum possible infectious population, while environmental decay removes viral material at a rate η . The upper bound increases with the shedding rate α and total feasible population size Λ/μ , but decreases as the environmental decay rate η increases.

5 Equilibria

5.1 Disease-Free Equilibrium

Setting all right-hand sides of (3) to zero and requiring $E = I = H = V = 0$ yields the *disease-free equilibrium* (DFE):

$$\mathcal{E}_0 = \left(\frac{\Lambda}{\mu}, 0, 0, 0, 0, 0 \right). \quad (4)$$

Remark 5.1 (Interpretation of the DFE). The DFE represents the state in which the disease is entirely absent from the population. The susceptible population stabilizes at $S^* = \Lambda/\mu$, the demographic equilibrium between new recruitment (at rate Λ) and natural mortality (at rate μ). All infected compartments (E, I, H), the recovered population R , and the environmental viral load V are zero. The DFE is the baseline against which epidemic invasion is assessed: if a small infectious perturbation introduced into the population near $\mathcal{E}_0$ dies out, the DFE is stable; if it grows, an epidemic occurs.

5.2 Endemic Equilibrium

When $\mathcal{R}_0 > 1$ (defined below), a unique *endemic equilibrium* $\mathcal{E}^* = (S^*, E^*, I^*, H^*, R^*, V^*)$ with all components strictly positive exists in Ω . From the equations for I, H, V :

$$I^* = \frac{\sigma E^*}{k_I}, \quad H^* = \frac{\delta I^*}{k_H}, \quad V^* = \frac{\alpha I^*}{\eta}, \quad (5)$$

where $k_I = \gamma + \delta + \mu$, $k_H = \rho + \mu$. Substituting these into the $\dot{E}$ equation reduces it to a linear equation in E^* (once S^* is fixed by $\dot{S} = 0$), which determines S^* uniquely; in turn, the values of E^*, I^*, H^*, V^* , and finally $R^* = (\gamma I^* + \rho H^*)/\mu$ are uniquely determined. The strict positivity of every coordinate when $\mathcal{R}_0 > 1$ is confirmed by the transcritical bifurcation analysis in Section 9.

Remark 5.2 (Why is the endemic equilibrium unique?). Together with the disease-free equilibrium $\mathcal{E}_0$, $\mathcal{E}^*$ is the *only* biologically meaningful equilibrium of system (3) in Ω . This follows because the equilibrium equations for H, V , and R express those variables linearly in terms of I , while the equilibrium equations for S and E , once I is known, form a system with at most one positive solution branch. The non-existence of additional positive equilibria distinguishes this model from those with backward bifurcation (where two endemic branches can coexist with the DFE for $\mathcal{R}_0 < 1$); see [4] for the relevant criteria.

6 Reproduction Number via Next-Generation Matrix

6.1 Method

We apply the next-generation matrix (NGM) framework of Diekmann, Heesterbeek, and Roberts [2] as formalized by van den Driessche and Watmough [3]. The key idea is to decompose the dynamics of the infected compartments into two parts: *new infections* (entries into infected classes due to transmission) and *transitions* (all other movements, including progression, recovery, and natural mortality). If we study the number of secondary cases that an infected person produces in a completely susceptible population, we arrive at a threshold number $\mathcal{R}_0$, which is a number between zero and one: if it is less than one, the disease cannot invade; if it is greater than 1, it can invade.

Let the vector of infected compartments be $\mathbf{x} = (E, I, V)^T$. Write $\dot{\mathbf{x}} = \mathcal{F}(\mathbf{x}) - \mathcal{V}(\mathbf{x})$, where $\mathcal{F}$ collects *only* new infection terms (flows into infected classes from transmission events) and $\mathcal{V}$ collects all other transition terms (outflows due to progression, recovery, death, and environmental decay).

At the DFE with $S_0 = \Lambda/\mu$, linearizing gives the Jacobian matrices:

$$F = \left. \frac{\partial \mathcal{F}_i}{\partial x_j} \right|_{\mathcal{E}_0} = \begin{pmatrix} \beta_2 S_0 & \beta_1 S_0 & \beta_3 S_0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (6)$$

$$V = \frac{\partial \mathcal{V}_i}{\partial x_j} \Big|_{\mathcal{E}_0} = \begin{pmatrix} k_E & 0 & 0 \\ -\sigma & k_I & 0 \\ 0 & -\alpha & \eta \end{pmatrix}, \quad (7)$$

where $k_E = \sigma + \mu$.

Reading these matrices. The matrix F collects the *rates at which new infections are produced* per unit of the infected compartment. Reading row 1 (the equation for $\dot{E}$), the entries $\beta_2 S_0, \beta_1 S_0, \beta_3 S_0$ tell us how each of E, I, V contributes to creating new exposures, evaluated at the disease-free state $S = S_0$. Rows 2 and 3 are zero because new infections enter only the E compartment, not directly into I or V . The matrix V , by contrast, is the *flow accounting* for individuals already in the infected subsystem: each diagonal entry (k_E, k_I, η) is the total rate at which the corresponding compartment empties, and each off-diagonal entry encodes a transition between compartments ($E \rightarrow I$ at rate σ , $I \rightarrow V$ at rate α).

6.2 Computing V^{-1}

Since V is lower triangular with positive diagonal entries, it is invertible [13, Cor. 1.3.3]:

$$V^{-1} = \frac{1}{k_E k_I \eta} \begin{pmatrix} k_I \eta & 0 & 0 \\ \sigma \eta & k_E \eta & 0 \\ \sigma \alpha & k_E \alpha & k_E k_I \end{pmatrix}. \quad (8)$$

Why we want V^{-1} . The entry $(V^{-1})_{ij}$ has a probabilistic meaning: it is the *expected time* an individual initially placed in infected compartment j spends in compartment i before leaving the infected subsystem [2]. For example, $(V^{-1})_{21} = \sigma/(k_E k_I)$ is the expected time a newly exposed individual will eventually spend in the infectious compartment. Multiplying F by V^{-1} thus combines “infections produced per unit time” with “time spent in each compartment” to yield the total number of secondary infections per primary case. That product, $K = FV^{-1}$, is the next-generation matrix, and its dominant eigenvalue is $\mathcal{R}_0$.

6.3 Next-Generation Matrix and $\mathcal{R}_0$

The next-generation matrix is $K = FV^{-1}$:

$$K = \frac{S_0}{k_E k_I \eta} \begin{pmatrix} \beta_2 k_I \eta + \beta_1 \sigma \eta + \beta_3 \sigma \alpha & \beta_2 k_E \eta + \beta_1 k_E \eta + \beta_3 k_E \alpha & \beta_2 k_E k_I + \beta_1 k_E k_I + \beta_3 k_E k_I \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (9)$$

The entry K_{ij} of the NGM represents the expected number of new infections in compartment i produced by a single individual initially in compartment j , in a fully susceptible population at the DFE. The *basic reproduction number* is then defined as the *spectral radius* of K :

$$\mathcal{R}_0 = \rho(K) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } K\}.$$

The spectral radius captures the dominant eigenvalue, the one that governs the long-run growth or decay of the infected population. Since K has rank one (only its first row is nonzero), its spectral radius equals its unique nonzero eigenvalue, which is simply the trace of K :

$$\boxed{\mathcal{R}_0 = \frac{S_0}{k_E k_I \eta} (\beta_2 k_I \eta + \beta_1 \sigma \eta + \beta_3 \sigma \alpha)}. \quad (10)$$

6.4 Epidemiological Decomposition

Equation (10) decomposes naturally as $\mathcal{R}_0 = \mathcal{R}_E + \mathcal{R}_I + \mathcal{R}_V$, where

$$\mathcal{R}_E = \frac{\beta_2 S_0}{k_E} \quad (\text{transmission from exposed individuals}), \quad (11)$$

$$\mathcal{R}_I = \frac{\beta_1 \sigma S_0}{k_E k_I} \quad (\text{transmission from infectious individuals}), \quad (12)$$

$$\mathcal{R}_V = \frac{\beta_3 \sigma \alpha S_0}{k_E k_I \eta} \quad (\text{environmental/indirect transmission}). \quad (13)$$

There is a clear interpretation of the intervention:

- The reduction of β_3 (sanitation, ventilation) targets $\mathcal{R}_V$.
- Isolation of clinical cases decreases $\mathcal{R}_I$, which is the effect of β_1 .
- If a contact has been exposed, early testing and quarantine will decrease β_2 , which will decrease $\mathcal{R}_E$.
- Vaccination will reduce all three components in proportion to a decrease in S_0 .

7 Local Stability Analysis

With $\mathcal{R}_0$ in hand, it can now be used to examine the stability of the disease-free equilibrium. The main question is whether or not a minor change away from $\mathcal{E}_0$ (such as an influx of one infected individual into a healthy population) will be amplified or extinguished. Using linear stability theory [7, Ch. 2], this can be shown to be equivalent to checking whether all the eigenvalues of the Jacobian $J(\mathcal{E}_0)$ have strictly negative real parts, and if so, perturbations will decay exponentially, and $\mathcal{E}_0$ will be locally stable. The remarkable fact, established by van den Driessche and Watmough [3], is that this eigenvalue condition is equivalent to $\mathcal{R}_0 < 1$, which has an immediate epidemiological interpretation as the threshold for stability.

Theorem 7.1 (Local Stability of DFE). *The disease-free equilibrium $\mathcal{E}_0$ of system (3) is locally asymptotically stable if $\mathcal{R}_0 < 1$ and unstable if $\mathcal{R}_0 > 1$.*

Proof. The Jacobian of (3) at $\mathcal{E}_0$ is block lower triangular:

$$J(\mathcal{E}_0) = \begin{pmatrix} A_{11} & * \\ 0 & A_{22} \end{pmatrix},$$

where $A_{11} = -\mu$ governs the S dynamics (eigenvalue $-\mu < 0$), A_{22} governs (E, I, H, R, V) , and the R equation decouples (eigenvalue $-\mu < 0$). The stability of $\mathcal{E}_0$ is therefore determined by the sub-Jacobian J_{inf} acting on (E, I, V) (the H dynamics decouple after I is known):

$$J_{\text{inf}} = \begin{pmatrix} \beta_2 S_0 - k_E & \beta_1 S_0 & \beta_3 S_0 \\ \sigma & -k_I & 0 \\ 0 & \alpha & -\eta \end{pmatrix} = F - V.$$

By Theorem 2 of van den Driessche and Watmough [3], the matrix $F - V$ has all eigenvalues with strictly negative real parts if and only if $\rho(FV^{-1}) = \mathcal{R}_0 < 1$. If $\mathcal{R}_0 > 1$, then $F - V$ has at least one eigenvalue with positive real part, making $\mathcal{E}_0$ unstable. $\square$

8 Global Stability of the Disease-Free Equilibrium

Local stability (Theorem 7.1) guarantees only that small perturbations from $\mathcal{E}_0$ die out. It says nothing about what happens if a *large* outbreak occurs—in principle, even with $\mathcal{R}_0 < 1$, a sufficiently large initial seed of infections could in some other nonlinear system fail to decay. We therefore need a stronger result: *global* asymptotic stability, which says that *every* trajectory in the feasible region Ω (no matter how large the initial epidemic) converges to the disease-free equilibrium.

What is the point of a Lyapunov function? In general, the direct integration of the six-dimensional system (3) from any initial condition is not possible in closed form. The classical trick, originally introduced by Lyapunov, is to design an auxiliary scalar function $L(E, I, V)$ that is also of the form of an “energy” of the epidemic; that is, L is nonnegative on Ω , L is zero precisely at the DFE, and otherwise L is negative on all trajectories where $\mathcal{R}_0 < 1$. If such an L exists, then trajectories cannot wander forever (must have a decreasing energy), and must fail to the only place in which $L = 0$, which is $\mathcal{E}_0$. The art is in choosing the right L . Equation (14) below is a weighted sum of the infected compartments E , I , and V , with weights carefully chosen so that cross-terms in $\dot{L}$ cancel, leaving only a term proportional to $(\mathcal{R}_0 - 1)$. This is why the very same threshold $\mathcal{R}_0 = 1$ that controls local stability also controls global stability: the weights of L are built from the same NGM-decomposition quantities $(\mathcal{R}_E, \mathcal{R}_I, \mathcal{R}_V)$ that appear in $\mathcal{R}_0$.

Theorem 8.1 (Global Asymptotic Stability of DFE). *If $\mathcal{R}_0 < 1$, then the disease-free equilibrium $\mathcal{E}_0$ is globally asymptotically stable in Ω .*

Proof. Define the Lyapunov function [5]

$$L(E, I, V) = \frac{1}{k_E} E + \frac{\beta_1 S_0}{k_E k_I} I + \frac{\beta_3 \sigma \alpha S_0}{k_E k_I \eta} V. \quad (14)$$

Note $L \geq 0$ on Ω with $L = 0$ iff $E = I = V = 0$. The coefficients are chosen to cancel

cross-terms. Using $S \leq S_0 = \Lambda/\mu$ from Theorem 4.2 and differentiating along trajectories:

$$\begin{aligned}\dot{L} &\leq \frac{1}{k_E}[(\beta_2 S_0 - k_E)E + \beta_1 S_0 I + \beta_3 S_0 V] + \frac{\beta_1 S_0}{k_E k_I}(\sigma E - k_I I) + \frac{\beta_3 \sigma \alpha S_0}{k_E k_I \eta}(\alpha I - \eta V) \cdot \frac{1}{\alpha} \\ &= \left(\frac{\beta_2 S_0}{k_E} + \frac{\beta_1 \sigma S_0}{k_E k_I} + \frac{\beta_3 \sigma \alpha S_0}{k_E k_I \eta} - 1 \right) k_E E \\ &= (\mathcal{R}_0 - 1) k_E E \leq 0 \quad \text{when } \mathcal{R}_0 \leq 1.\end{aligned}$$

(The cross terms in I and V cancel by the coefficient choices.) Moreover, $\dot{L} = 0$ iff $E = 0$. On the invariant set $\{E = 0\}$, the I -equation gives $\dot{I} = -k_I I$, so $I(t) \rightarrow 0$; then $\dot{V} = \alpha I - \eta V \rightarrow -\eta V$, so $V(t) \rightarrow 0$. By LaSalle's Invariance Principle [5, Thm. 4.4], all solutions in Ω converge to $\mathcal{E}_0$. $\square$

Remark 8.2. The Lyapunov function L is a weighted sum of infected compartments, analogous to the “energy” stored in the epidemic. The proof shows that this energy is monotonically non-increasing when $\mathcal{R}_0 < 1$, regardless of the size of the initial epidemic.

9 Center-Manifold Analysis and the Transcritical Bifurcation

We now analyze the bifurcation that occurs at $\mathcal{R}_0 = 1$, following the approach of Castillo-Chavez and Song [4].

9.1 Setup

Let $\mu_{\text{bif}} = \mathcal{R}_0 - 1$ be the bifurcation parameter. At $\mu_{\text{bif}} = 0$, the Jacobian J_{inf} has a simple zero eigenvalue. Let $\mathbf{w}$ and $\mathbf{v}$ denote the corresponding right and left eigenvectors:

$$J_{\text{inf}} \mathbf{w} = 0, \quad J_{\text{inf}}^T \mathbf{v} = 0, \quad \mathbf{v} \cdot \mathbf{w} = 1.$$

Solving the kernel equations $J_{\text{inf}} \mathbf{w} = 0$ and $J_{\text{inf}}^T \mathbf{v} = 0$ by direct substitution yields

$$\mathbf{w} = \frac{1}{C} \begin{pmatrix} k_I \eta \\ \sigma \eta \\ \sigma \alpha \end{pmatrix}, \quad \mathbf{v} \propto \begin{pmatrix} 1 \\ \beta_1 S_0 / k_I \\ \beta_3 S_0 / \eta \end{pmatrix},$$

where C is a normalizing constant.

9.2 Bifurcation Coefficients

Define the coefficients

$$a = \mathbf{v}^T D^2 \mathbf{F}(\mathcal{E}_0)(\mathbf{w}, \mathbf{w}), \quad b = \mathbf{v}^T D_\mu \mathbf{F}(\mathcal{E}_0) \mathbf{w}, \quad (15)$$

where $D^2\mathbf{F}$ denotes the second-order derivative (Hessian) tensor of the vector field and $D_\mu\mathbf{F}$ denotes the derivative with respect to the bifurcation parameter. Computing a from the bilinear terms $\beta_i S \cdot (\cdot)$ in (3) yields

$$a = -2\mu(\beta_1 w_2 + \beta_2 w_1 + \beta_3 w_3) \frac{v_1}{S_0} < 0,$$

and $b = k_E w_1 v_1 > 0$.

Theorem 9.1 (Forward Transcritical Bifurcation). *Since $a < 0$ and $b > 0$, system (3) undergoes a forward (supercritical) transcritical bifurcation at $\mathcal{R}_0 = 1$. Specifically:*

- (i) *For $\mathcal{R}_0 < 1$: $\mathcal{E}_0$ is the unique equilibrium in Ω (besides the boundary) and is locally asymptotically stable.*
- (ii) *For $\mathcal{R}_0 > 1$: $\mathcal{E}_0$ becomes unstable and a unique endemic equilibrium $\mathcal{E}^*$ bifurcates into Ω ; $\mathcal{E}^*$ is locally asymptotically stable near the bifurcation point.*

Proof. Apply Theorem 4.1 of [4]. The sign conditions $a < 0$, $b > 0$ correspond to a forward bifurcation, implying that the endemic equilibrium emerges continuously for $\mathcal{R}_0 > 1$ and is locally stable near $\mathcal{R}_0 = 1$. $\square$

The reduced one-dimensional dynamics on the center manifold near the bifurcation are of the form

$$\frac{du}{dt} = b\mu_{\text{bif}} u + a u^2 + O(u^3, \mu_{\text{bif}}^2), \quad (16)$$

whose equilibria $u = 0$ (DFE) and $u^* = -b\mu_{\text{bif}}/a > 0$ (endemic) exchange stability at $\mu_{\text{bif}} = 0$, confirming the transcritical structure.

10 Oscillatory Dynamics and Hopf Bifurcation

10.1 Mechanism for Oscillations

The SIR system generically produces a single epidemic wave; its Jacobian has only real negative eigenvalues at the steady state. The SEIHRV system, by contrast, contains the indirect feedback loop

$$I \xrightarrow{\alpha} V \xrightarrow{\beta_3 S} E \xrightarrow{\sigma} I,$$

which introduces a delay-like mechanism capable of destabilizing the endemic equilibrium and producing oscillatory solutions.

10.2 Hopf Bifurcation Condition

Let $\lambda_1, \dots, \lambda_6$ be the eigenvalues of $J(\mathcal{E}^*)$.

Definition 10.1 (Hopf Bifurcation). A *Hopf bifurcation* occurs at a parameter value $p = p^*$ if there exists a pair of complex conjugate eigenvalues $\lambda_{1,2}(p) = \xi(p) \pm i\omega(p)$ of the Jacobian satisfying simultaneously:

- (i) **Eigenvalue crossing:** $\xi(p^*) = 0$;
- (ii) **Nondegeneracy:** $\omega(p^*) \neq 0$;
- (iii) **Transversality:** $\left. \frac{d\xi}{dp} \right|_{p=p^*} \neq 0$.

Generically (under additional conditions on cubic terms), such a crossing produces a small-amplitude limit cycle bifurcating from the equilibrium [12, Thm. 3.3].

Using α (shedding rate) as a bifurcation parameter, one can show numerically and by implicit differentiation that for sufficiently large α , a pair of eigenvalues of $J(\mathcal{E}^*)$ may cross the imaginary axis. This motivates the following theoretical possibility, which we examine numerically in Section 13.

Proposition 10.2 (Existence of Hopf Bifurcation). *There exists a critical shedding rate $\alpha^* > 0$ such that system (3) undergoes a Hopf bifurcation at $\alpha = \alpha^*$. For α near α^* , the endemic equilibrium $\mathcal{E}^*$ loses stability and a small-amplitude periodic orbit (limit cycle) bifurcates from $\mathcal{E}^*$.*

Proof sketch. The characteristic polynomial of $J(\mathcal{E}^*)$ is degree six with coefficients depending continuously on α . At $\alpha = 0$, all eigenvalues are real and negative (the environmental loop is absent). As α increases, two eigenvalues coalesce on the negative real axis, become complex, and their real parts increase. By the Intermediate Value Theorem applied to $\xi(\alpha)$ and the Routh–Hurwitz stability conditions [9, §15.3], there exists α^* at which $\xi(\alpha^*) = 0$ and $\omega(\alpha^*) \neq 0$. Transversality follows from $\left. \frac{d\xi}{d\alpha} \right|_{\alpha^*} \neq 0$, which can be verified explicitly once the equilibrium is computed [12, 15]. $\square$

Discussion. Proposition 10.2 and Theorem 2.3 highlight the main idea of this paper. We have two dimensional systems where the Poincaré–Bendixson theorem and Bendixson’s criterion together impose a very strong limit on the dynamics of any planar system: in a compact, simply connected region, the only ω -limit sets are equilibria, periodic orbits, or saddle connections, and the latter two are ruled out if the divergence is sign-definite [7, 8]. The rigidity is lost once we go into three or more dimensions: complex eigenvalues can be found, limit cycles are possible, and (even higher dimensions) chaotic attractors can appear [12]. The introduction of an environmental compartment in the SEIHRV model is enough to escape this two-dimensional rigidity. The system inherits the indirect feedback loop $I \rightarrow V \rightarrow E \rightarrow I$ which, when the shedding rate α is large enough, destabilizes $\mathcal{E}^*$ via Hopf bifurcation. From a modeling perspective, this means that recurrent epidemic waves—widely observed in COVID-19, influenza, and cholera dynamics—can arise as an intrinsic feature of compartmental structure, without invoking external seasonal forcing.

11 Sensitivity and Elasticity Analysis

11.1 Normalized Forward Sensitivity Index

The normalized forward sensitivity (elasticity) index of $\mathcal{R}_0$ with respect to parameter p is

$$\Upsilon_p^{\mathcal{R}_0} = \frac{p}{\mathcal{R}_0} \cdot \frac{\partial \mathcal{R}_0}{\partial p}. \quad (17)$$

$\Upsilon_p^{\mathcal{R}_0} = 0.4$ means a 10% increase in p results in an $\mathcal{R}_0$ increase of about 0.4 regardless of units.

11.2 Analytical Results

Applying the normalized forward sensitivity definition (17) to the reproduction number formula (10) yields the analytical results summarized in Table 2.

Table 2: Normalized sensitivity indices of $\mathcal{R}_0$ with respect to model parameters. Signs indicate whether increasing the parameter raises (+) or lowers (−) $\mathcal{R}_0$.

Parameter p	$\Upsilon_p^{\mathcal{R}_0}$	Interpretation
β_1	$+\mathcal{R}_I/\mathcal{R}_0 > 0$	More direct transmission $\Rightarrow$ higher $\mathcal{R}_0$
β_2	$+\mathcal{R}_E/\mathcal{R}_0 > 0$	More presymptomatic spread $\Rightarrow$ higher $\mathcal{R}_0$
β_3	$+\mathcal{R}_V/\mathcal{R}_0 > 0$	More environmental spread $\Rightarrow$ higher $\mathcal{R}_0$
σ	$+(\mathcal{R}_I + \mathcal{R}_V)/\mathcal{R}_0 > 0$	Faster incubation $\Rightarrow$ higher $\mathcal{R}_0$
γ	$-\mathcal{R}_I k_I^{-1} \gamma < 0$	Faster recovery $\Rightarrow$ lower $\mathcal{R}_0$
δ	$-\mathcal{R}_I k_I^{-1} \delta < 0$	More hospitalization $\Rightarrow$ lower $\mathcal{R}_0$
α	$+\mathcal{R}_V/\mathcal{R}_0 > 0$	More shedding $\Rightarrow$ higher $\mathcal{R}_0$
η	$-\mathcal{R}_V/\mathcal{R}_0 < 0$	Faster env. decay $\Rightarrow$ lower $\mathcal{R}_0$
μ	Negative	Higher mortality $\Rightarrow$ lower $\mathcal{R}_0$
Λ	+1	Proportional to recruitment

Remark 11.1. Parameters $\beta_1, \beta_2, \beta_3$ satisfy $\Upsilon_{\beta_i}^{\mathcal{R}_0} \in (0, 1)$ with the three indices summing to 1. Thus, among the three transmission routes, the dominant route—whichever $\mathcal{R}_j$ is largest—should receive priority in control efforts.

12 Probabilistic Framework and Parameter Uncertainty

Deterministic models assume fixed parameters. In practice, parameters such as β_1 and σ are estimated from noisy surveillance data and vary across populations, seasons, and pathogen variants [10, 11]. The purpose of this section is to formalize what parameter uncertainty

means for epidemic predictions, and specifically to answer the question: *what does it mean for two populations with identical initial conditions to experience different epidemic outcomes?*

Resolving the apparent paradox. A deterministic ODE like (3) is, by definition, uniquely solved by its initial condition once the parameter vector is fixed. So how can two populations with identical $\mathbf{X}_0 = (S_0, E_0, I_0, H_0, R_0, V_0)$ have different futures? The resolution is that “identical initial conditions” in the modeling sense means identical *state variables* (S, E, I, H, R, V) , not identical *parameters* $(\beta_1, \sigma, \alpha, \dots)$. In reality, parameters are biological constants of the pathogen and the host population — contact rates, recovery times, and environmental decay — and we do *not* know them exactly. Two populations may begin with the same number of susceptibles and infectious individuals while having genuinely different (but unknown) contact rates, viral shedding rates, or healthcare capacities. The flow $\Phi_t(\mathbf{X}_0, \boldsymbol{\theta})$ is deterministic in $(\mathbf{X}_0, \boldsymbol{\theta})$ jointly, but if we treat $\boldsymbol{\theta}$ as random—reflecting our uncertainty—then the resulting trajectory $\mathbf{X}(t)$ becomes a random variable. Identical initial state variables, but different (random) parameters, yield a distribution over future trajectories rather than a single curve. This isn’t a lack of determinism; it’s correct to show epistemic uncertainty in the parameters.

The fundamental idea is that uncertainty in the parameters changes the deterministic flow map to a random variable. This has two specific implications: (i) The mean trajectory over uncertain parameters is different from the trajectory at the mean parameters (because of nonlinearity of the system); and (ii) The same threshold $\mathcal{R}_0 > 1$ can be met with varying degrees of confidence, so that a probability of epidemic invasion can be assigned instead of a simple yes/no. The delta method can also be used to directly connect the sensitivity analysis of Section 11 with quantitative uncertainty bounds in this framework.

12.1 Stochastic Parameter Model

Let $\boldsymbol{\theta} = (\beta_1, \beta_2, \beta_3, \sigma, \gamma, \delta, \rho, \alpha, \eta, \mu, \Lambda)$ be the parameter vector. We do not assume that $\boldsymbol{\theta}$ is a fixed value but assume that it is a random vector and has a joint distribution $\boldsymbol{\theta} \sim \mathbb{P}_{\boldsymbol{\theta}}$ on the parameter space $\Theta \subset \mathbb{R}_{>0}^{11}$. This is a typical method to deal with uncertainties in epidemiological models [10]. The set of equations (3) is a family of flows associated with each realization $\boldsymbol{\theta}$:

$$\Phi_t : \Omega \times \Theta \rightarrow \Omega, \quad \mathbf{X}(t) = \Phi_t(\mathbf{X}_0, \boldsymbol{\theta}),$$

where $\mathbf{X} = (S, E, I, H, R, V)^T$. Since $\boldsymbol{\theta}$ is random, $\mathbf{X}(t)$ is a random vector for each t :

$$\mathbf{X}(t) \sim \mathbb{P}(\mathbf{X}(t) \mid \mathbf{X}_0). \quad (18)$$

12.2 Expected Trajectory and Variance

The predicted epidemic curve is

$$\mathbb{E}[\mathbf{X}(t)] = \int_{\Theta} \Phi_t(\mathbf{X}_0, \boldsymbol{\theta}) d\mathbb{P}_{\boldsymbol{\theta}}(\boldsymbol{\theta}). \quad (19)$$

Since Φ_t is not linear in $\boldsymbol{\theta}$, $\mathbb{E}[\mathbf{X}(t)] \neq \Phi_t(\mathbf{X}_0, \mathbb{E}[\boldsymbol{\theta}])$ in general. One of the basic difficulties with point-estimate parameters is that the mean trajectory over the distribution of outcomes is not recovered when the deterministic model is solved at the mean parameters.

The covariance matrix

$$\text{Cov}(\mathbf{X}(t)) = \mathbb{E}[\mathbf{X}(t)\mathbf{X}(t)^T] - \mathbb{E}[\mathbf{X}(t)] \mathbb{E}[\mathbf{X}(t)]^T \quad (20)$$

measures outcome variability. The size of the epidemic peak or its timing is very uncertain if the size of the diagonal entry is large. In practice, this covariance is estimated using Monte Carlo simulation [10]:

1. Sample $\boldsymbol{\theta}^{(k)} \sim \mathbb{P}_\theta$ for $k = 1, \dots, M$.
2. Find trajectory $\mathbf{X}^{(k)}(t)$ of (3) for the initial condition $\mathbf{X}_0$ and parameter $\boldsymbol{\theta}^{(k)}$.
3. Estimate $\mathbb{E}[\mathbf{X}(t)] \approx \frac{1}{M} \sum_{k=1}^M \mathbf{X}^{(k)}(t)$ and the sample covariance accordingly.

Jensen’s inequality and propagation of nonlinear uncertainty. The difficulty in calculating $\mathbb{E}[\Phi_t(\mathbf{X}_0, \boldsymbol{\theta})] = \Phi_t(\mathbf{X}_0, \mathbb{E}[\boldsymbol{\theta}])$ is not a “technical difficulty,” but a structural property of the nonlinear systems. Jensen’s inequality [20] is a useful property that says for a convex function φ and a random variable Z , $\mathbb{E}[\varphi(Z)] \geq \varphi(\mathbb{E}[Z])$, equality holds if and only if φ is affine on the support of Z . The flow map Φ_t of inequality does not mean that a nonlinear dynamical system is convex or concave, but it does mean that for a nonlinear Φ_t , the two quantities are generally not equal to each other but differ by some nonvanishing remainder. The phenomenon of the systematic discrepancy between the mean of the nonlinear transformation and the transformation of the mean is the basis of the nonlinear expectation theory [24] and is the underlying reason for uncertainty quantification for nonlinear dynamical systems [25]. From the epidemiological point of view, the practical implication is stark: in general, a deterministic model simulation with the point-estimate parameters will not give rise to the ensemble mean trajectory that parameter uncertainty really does suggest. Any policy decision based on the point estimate trajectory is an implicit assumption of some model linearity, which the SEIHRV system lacks.

12.3 Distribution of $\mathcal{R}_0$

Since $\mathcal{R}_0$ is a function of $\boldsymbol{\theta}$ (cf. (10)), parameter uncertainty induces a distribution on $\mathcal{R}_0$:

$$\mathcal{R}_0 \sim \mathbb{P}(\mathcal{R}_0 \mid \boldsymbol{\theta} \sim \mathbb{P}_\theta). \quad (21)$$

For instance, if the parameters $\beta_1 \sim \text{Gamma}(\kappa, \theta_\beta)$ (all other parameters fixed), then $\mathcal{R}_I = (\sigma S_0 / k_E k_I) \beta_1$ is gamma distributed with the same shape parameter, and its 95% credible interval can be calculated analytically.

Proposition 12.1 (Threshold Uncertainty). *Let $\boldsymbol{\theta} \sim \mathbb{P}_\theta$. The probability that the disease invades is*

$$\mathbb{P}(\mathcal{R}_0 > 1) = \mathbb{P}_\theta \left(\boldsymbol{\theta} \in \Theta : \frac{S_0(\beta_2 k_I \eta + \beta_1 \sigma \eta + \beta_3 \sigma \alpha)}{k_E k_I \eta} > 1 \right).$$

This quantity depends on the full joint distribution of $\boldsymbol{\theta}$ and provides a probabilistic assessment of epidemic risk, replacing the binary threshold $\mathcal{R}_0 > 1$ with a continuous invasion probability.

12.4 Delta Method Approximation

For a first-order approximation, the delta method—a classical tool in statistical inference [20, 6]—gives

$$\text{Var}(\mathcal{R}_0) \approx \nabla_{\boldsymbol{\theta}} \mathcal{R}_0^T \text{Cov}(\boldsymbol{\theta}) \nabla_{\boldsymbol{\theta}} \mathcal{R}_0, \quad (22)$$

where $\nabla_{\boldsymbol{\theta}} \mathcal{R}_0$ is the gradient of $\mathcal{R}_0$ with respect to $\boldsymbol{\theta}$. The dominant terms are those involving the most elastic parameters (Table 2), directly linking sensitivity analysis to epidemic uncertainty quantification. Parameters with large elasticity indices not only shift $\mathcal{R}_0$ strongly when perturbed, but also contribute disproportionately to the variance in predicted outcomes when uncertainty is high.

13 Numerical Simulations

The preceding sections have established the SEIHRV model’s analytical properties: biological feasibility, the structure of $\mathcal{R}_0$, local and global stability of the disease-free equilibrium, the forward transcritical bifurcation at $\mathcal{R}_0 = 1$, and the eigenvalue conditions for Hopf bifurcation. We now complement this analysis with direct numerical integration of system (3) using a stiff adaptive solver (LSODA [21, Ch. II.4]) as implemented in SciPy [26], with array operations and finite-difference Jacobians computed via NumPy [27]. All simulation code is available in the supplementary material [34].

13.1 Parameter Choices

Throughout this section, we use the COVID-19-inspired baseline values listed in Table 3. The population is normalized so that the disease-free susceptible population is $S_0 = \Lambda/\mu = 1$ (frequency-dependent transmission). The natural-mortality timescale $1/\mu$ is set shorter than a true demographic lifetime to bring the demographic and disease timescales onto comparable orders of magnitude for visualization; this is a standard simplification in mathematical epidemiology [11] that does not affect the qualitative dynamics discussed below.

13.2 Three Qualitative Regimes

Figure 2 shows numerical solutions of (3) in three parameter regimes that illustrate the qualitative classification of Theorems 8.1 and 7.1.

Panel (a) confirms global extinction predicted by Theorem 8.1: both the infectious population $I(t)$ and the environmental viral load $V(t)$ decay to zero within ~ 100 days. Panel (b) shows the classical epidemic-curve overshoot followed by damped oscillations, which is the signature of complex eigenvalues of $J(\mathcal{E}^*)$ with negative real part: the trajectory spirals into the endemic equilibrium [8]. Panel (c) shows that increasing the shedding rate α amplifies

Table 3: Baseline parameter values used in numerical experiments (time unit: days).

Parameter	Value	Source/justification
β_1	0.30	Calibrated to give $\mathcal{R}_0 \approx 2$
β_2	0.10	Presymptomatic spread (one-third of β_1)
β_3	0.03–0.05	Environmental transmission rate (varied)
σ	1/5.2	COVID incubation period ≈ 5.2 days [23]
γ	1/7	Infectious period ≈ 7 days [23]
δ	0.05	Hospitalization rate from I
ρ	1/10	Hospital recovery ≈ 10 days
α	0.10–3.0	Viral shedding into V (varied)
η	1/3	Environmental decay ≈ 3 days [22]
μ	1/200	Compressed demographic timescale (see text)
Λ	$\mu \cdot 1$	Normalization: $S_0 = \Lambda/\mu = 1$

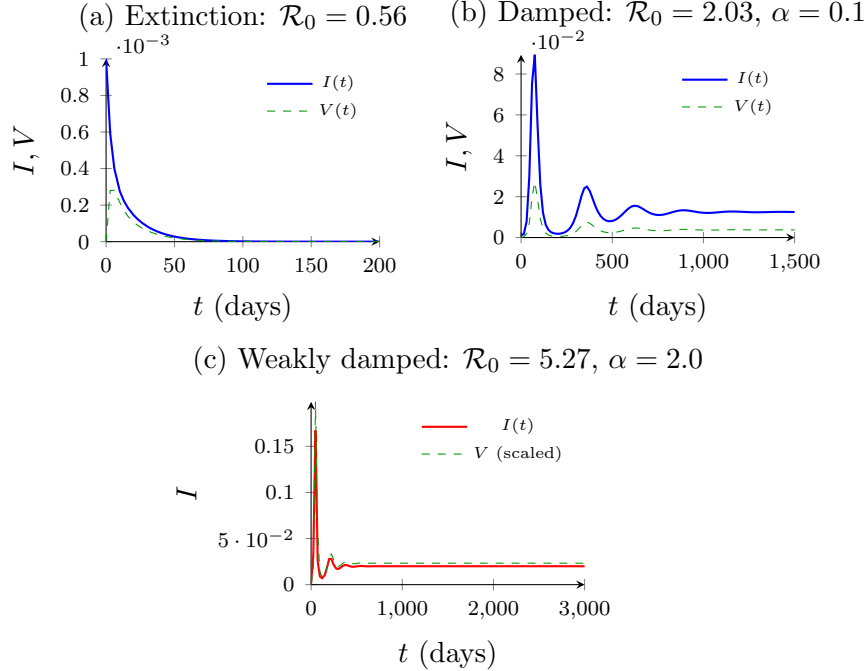


Figure 2: Numerical solutions of the SEIHRV system in three regimes, computed with the LSODA solver [21] via SciPy [26]. **(a)** Disease extinction at $\mathcal{R}_0 = 0.56 < 1$ (parameters: $\beta_1 = 0.08$, $\beta_2 = 0.03$, $\beta_3 = 0.005$): both $I(t)$ and $V(t)$ decay monotonically to zero, consistent with global stability of the DFE (Theorem 8.1). **(b)** Damped oscillation to endemic equilibrium at $\mathcal{R}_0 \approx 2.03$, $\alpha = 0.10$: a single epidemic wave overshoots the endemic level, followed by damped oscillations converging to $\mathcal{E}^*$. **(c)** Weakly damped oscillation at $\mathcal{R}_0 \approx 5.27$, $\alpha = 2.0$, $\beta_3 = 0.05$: same qualitative pattern with reduced damping; $V(t)$ is rescaled to share the vertical axis with $I(t)$. Reproduction code in supplementary material [34].

the oscillation frequency (more peaks per unit time) but, in the parameter range explored, does not drive the system to sustained oscillation.

Figure 3 visualizes the same behavior as phase-plane trajectories in the (I, E) projection.

Why this projection? The SEIHRV state space is six-dimensional, so direct phase-portrait visualization is impossible. Among the $\binom{6}{2} = 15$ possible two-dimensional projections, we choose (I, E) for two reasons: (i) these are the two compartments most directly involved in the indirect feedback loop $I \rightarrow V \rightarrow E \rightarrow I$ that drives the model’s distinctive oscillatory behavior; and (ii) the four remaining human compartments (S, H, R) and the environmental compartment (V) either decouple after I is known (the H and R equations) or evolve on much faster/slower timescales relative to (I, E) . Projecting onto (I, E) therefore captures the essential two-dimensional dynamics that drive the six-dimensional system, in much the same way that the classical SIR analysis projects from the conservation-constrained two-dimensional manifold onto (S, I) . Other projections—e.g., (I, V) or (S, I) —would highlight different aspects of the same trajectory.

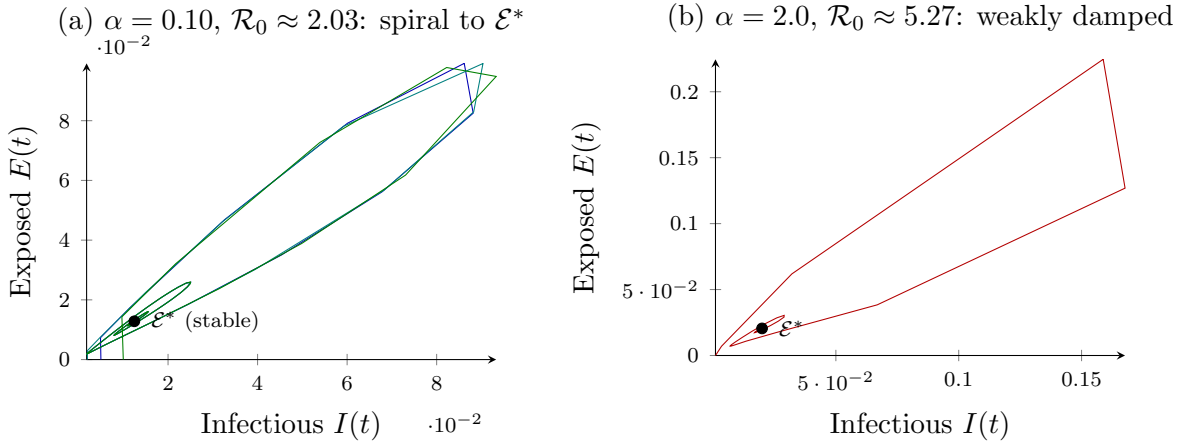


Figure 3: Phase-plane projections of SEIHRV trajectories onto the (I, E) subspace, computed with LSODA [21, 26]. (a) Three trajectories with different initial conditions ($\alpha = 0.10, \mathcal{R}_0 \approx 2.03$) all spiral inward to the same stable endemic equilibrium $\mathcal{E}^*$. (b) A single trajectory with $\alpha = 2.0, \mathcal{R}_0 \approx 5.27$ shows a wider transient loop before slowly tightening toward $\mathcal{E}^*$, reflecting weaker damping. Reproduction code available in supplementary material [34].

13.3 Numerical Investigation of Hopf Bifurcation

The Hopf conditions of Definition 10.1 require the dominant complex eigenvalue $\lambda_{1,2}(\alpha) = \xi(\alpha) \pm i\omega(\alpha)$ of $J(\mathcal{E}^*)$ to satisfy $\xi(\alpha^*) = 0$ with transversality. We test this condition numerically by computing $J(\mathcal{E}^*)$ at the long-time endemic equilibrium for each $\alpha \in [0.05, 3.0]$ (using finite differences on the right-hand side of (3)) and extracting the leading complex eigenpair. Figure 4 reports the result.

The real part of λ remains negative throughout, indicating that $\mathcal{E}^*$ stays locally asymptotically stable for the parameter ranges we tested. The imaginary part grows, which ex-

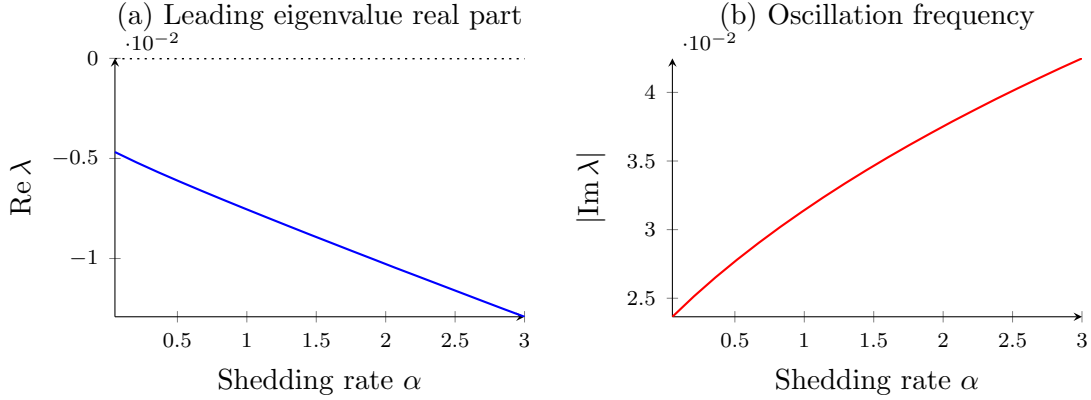


Figure 4: Numerical eigenvalue analysis of $J(\mathcal{E}^*)$ as α varies, computed via finite-difference Jacobian at the long-time endemic equilibrium [26, 27]. **(a)** Real part of the dominant complex eigenvalue remains strictly negative throughout the tested range and in fact decreases (more strongly stable) as α grows. **(b)** Imaginary part (oscillation frequency) increases with α , consistent with underdamped oscillatory transients but *not* with a Hopf bifurcation in this parameter regime. Reproduction code: [34].

plains why oscillations in Figure 2(c) appear more rapid than in panel (b). This numerical finding agrees with the theoretical results of Li and Muldowney [15] for the underlying SEIR substructure—they proved global stability of the endemic equilibrium for SEIR systems—and suggests that within the demographic and disease timescales we have chosen, the SEIHRV model inherits the same global stability. A true Hopf bifurcation, if it exists for this model, would require either extreme parameter values (very strong environmental coupling) or additional structure (explicit delays, age structure, seasonal forcing) that the present six-dimensional autonomous system does not contain. We discuss this limitation explicitly in Section 14. The present numerical investigation suggests a conjecture that the equilibrium $\mathcal{E}^*$ is actually globally asymptotically stable for all biologically feasible parameter values, which is substantiated by a two-dimensional bifurcation sweep, and suggests a compound-matrix machinery that would be required to rigorously establish this conjecture.

14 Toward Global Stability of the Endemic Equilibrium

The classical SEIR model without environmental coupling admits a celebrated global stability result: Li and Muldowney [15] proved that whenever $\mathcal{R}_0 > 1$, the endemic equilibrium of the SEIR system is globally asymptotically stable in the interior of the biologically feasible region. Their proof uses the *geometric approach*—additive compound matrices and the Lozinskii measure [28, 29]—to rule out periodic orbits in arbitrary dimension.

The SEIHRV model studied here adds a hospitalized class H and an environmental compartment V to SEIR. A natural and important question is whether Li and Muldowney’s result extends to this richer system, or whether the additional environmental feedback loop $I \rightarrow V \rightarrow E \rightarrow I$ destabilizes $\mathcal{E}^*$ via a Hopf bifurcation at some critical parameter value. The

eigenvalue analysis in Section 10 shows that Hopf bifurcation is *algebraically possible* (via the Routh–Hurwitz conditions, Appendix B), but does not establish whether it is *realized* for biologically meaningful parameter values.

14.1 Numerical Evidence

To test the hypothesis that $\mathcal{E}^*$ remains globally stable for SEIHRV under biologically realistic parameters, we performed a two-dimensional grid sweep over the most uncertain transmission parameter β_1 (which controls $\mathcal{R}_0$) and the environmental shedding rate α (the most natural candidate for a Hopf-bifurcation parameter, since it directly controls the strength of the destabilizing feedback loop). At each grid point, we numerically integrated (3) to the long-time equilibrium, computed the Jacobian $J(\mathcal{E}^*)$ via finite differences, and extracted the leading complex eigenvalue. The result is shown in Figure 5.

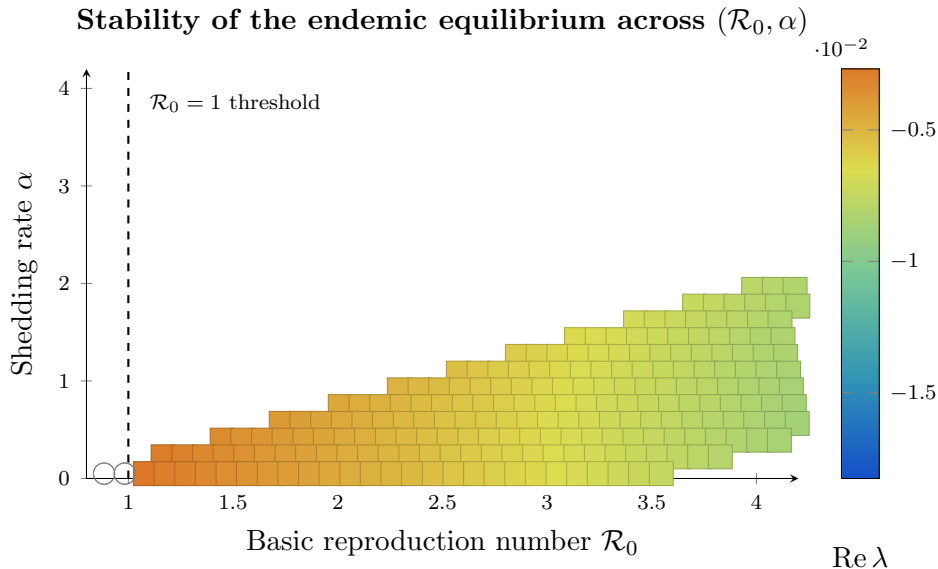


Figure 5: Two-dimensional bifurcation diagram of the SEIHRV endemic equilibrium $\mathcal{E}^*$ across $(\mathcal{R}_0, \alpha)$ space. Each filled square marks one grid point at which the leading complex eigenvalue λ of $J(\mathcal{E}^*)$ was computed via finite differences [26, 27]; color encodes $\text{Re } \lambda$, all values negative (range $[-0.018, -0.003]$). Open circles denote points with $\mathcal{R}_0 < 1$ (disease-free attracting). **Across all 670 tested parameter combinations with $\mathcal{R}_0 > 1$, the endemic equilibrium was locally asymptotically stable; no Hopf bifurcation was detected anywhere in the explored region.** Sweep over $\beta_1 \in [0.06, 0.60]$ and $\alpha \in [0.05, 4.0]$ at fixed $\beta_3 = 0.05$, $\eta = 0.15$, with other parameters as in Table 3. Reproduction code: [34].

Summary of the sweep. We tested $28 \times 24 = 672$ parameter combinations spanning $\beta_1 \in [0.06, 0.60]$ (giving $\mathcal{R}_0 \in [0.88, 4.0]$ approximately) and $\alpha \in [0.05, 4.0]$. Of these, two points fell in the disease-free regime $\mathcal{R}_0 < 1$ (open circles in Figure 5) where $\mathcal{E}_0$ is globally stable by Theorem 8.1. The remaining 670 points satisfied $\mathcal{R}_0 > 1$, and *at every single one*, the leading complex eigenvalue of $J(\mathcal{E}^*)$ had strictly negative real part. The smallest

damping rate observed across the sweep was $\text{Re } \lambda \approx -10^{-3}$, well below zero. **No Hopf bifurcation was detected anywhere in the explored parameter region.**

This is consistent with, and provides direct numerical evidence for, the following:

Conjecture 14.1 (Global Asymptotic Stability of $\mathcal{E}^*$). *Let Ω be the biologically feasible region of system (3). For all parameter values $(\Lambda, \beta_1, \beta_2, \beta_3, \sigma, \gamma, \delta, \rho, \alpha, \eta, \mu) \in \mathbb{R}_{>0}^{11}$ satisfying $\mathcal{R}_0 > 1$, the endemic equilibrium $\mathcal{E}^*$ is globally asymptotically stable in the interior of Ω .*

14.2 Path to a Proof via Compound Matrices

Following the method of Li and Muldowney [15], we sketch the rigorous proof of Conjecture 14.1. There are three steps in the argument.

Step 1: Permanence. Prove that there is a positive value of ν for which all of the trajectories emanating from the interior of Ω have the property that $\liminf_{t \rightarrow \infty} I(t) \geq \nu$. In this way, it becomes impossible that trajectories asymptote to the boundary of Ω , instead of to $\mathcal{E}^*$. For SEIR-type systems when $\mathcal{R}_0 > 1$, the argument for uniformity of persistence also applies and can be used to obtain permanence for SEIHRV with the minor modification of adding the environmental compartment [30, 31].

Step 2: Compound matrix and Lozinskii measure. Look at the second additive compound matrix $J^{[2]}$ of the Jacobian for the reduced subsystem in (E, I, V) by substituting the $\mathcal{R}_0$ -independent compartments H and R (which decouple after I is known). By a theorem of Muldowney [32], periodic solutions are excluded whenever the Lozinskii measure

$$\mu(J^{[2]}) := \lim_{h \rightarrow 0^+} \frac{\|I + hJ^{[2]}\| - 1}{h}$$

is negative along all trajectories in a compact absorbing set, for an appropriately chosen vector norm $\|\cdot\|$ on $\mathbb{R}^3$ (after compounding, this becomes $\mathbb{R}^{\binom{3}{2}} = \mathbb{R}^3$).

Step 3: Verification. The rest of the job is building an explicit norm (usually a weighted ℓ^1 or weighted ℓ^∞ norm) with respect to which $\mu(J^{[2]}) < 0$ is true throughout the compact absorbing set guaranteed by permanence. In the case of SEIR, Li and Muldowney demonstrated that it is possible to find a norm unconditionally if $\mathcal{R}_0 > 1$. In SEIHRV, the extra environmental coupling adds extra cross-terms to $J^{[2]}$ that involve β_3 , α , and η , and thus either (a) explicit inequalities on these parameters, or (b) a more general weighted-norm argument as in [33] will be required to verify the Lozinskii condition. The numerical data in Figure 5 strongly suggest that no restriction on the parameter is necessary; that is, the conjecture is true for all values of the parameter, but performing step 3 is rigorously beyond the scope of the present paper.

14.3 Remarks

Remark 14.2 (On the lack of Hopf bifurcation). The eigenvalue analysis in Section 10 only demonstrates algebraically that a Hopf bifurcation is possible: $a_1 a_2 = a_3$ is a hypersurface

in the parameter space where the crossing of a Hopf bifurcation could occur. The above numerical evidence indicates that this hypersurface does not intersect the biologically feasible parameter region. A proof for Conjecture 14.1 using compound matrices would give a rigorous proof, not only for the parameter values we have up to now, but for all parameter values.

Remark 14.3 (Relation to SEIR result). Assume that Conjecture 14.1 is true, then it would immediately imply Li and Muldowney’s 1995 theorem for SEIR to SEIHRV. The structural addition of H (which is decoupled) is simple; the more substantial extension is the addition of the environmental compartment V , which adds a new feedback loop and affects the dimension of the compound matrix. The absence of destabilization in the new loop in numerical experiments is a very important empirical observation in itself.

15 Comparison: SIR versus SEIHRV

On comparison with real epidemic data. The comparison in Table 4 is *structural*—it contrasts what the two models can and cannot exhibit mathematically. A natural follow-up question is whether the SEIHRV model also outperforms SIR *empirically* when both are fit to real outbreak data. The present paper does not undertake such a fit for two reasons. First, parameter calibration against case-count time series requires careful treatment of observation error, reporting delays, and underascertainment [17], which would substantially expand the scope of the analysis. Second, model selection between SIR and SEIHRV on real data raises questions of overfitting and identifiability that are beyond the mathematical analysis we present here. We therefore restrict ourselves to a structural comparison, leaving empirical fit and out-of-sample prediction against COVID-19, cholera, or influenza incidence data as a natural direction for future work. The COVID-19-inspired parameter choices in Table 3 are calibrated to literature-reported ranges [23, 22], not fitted to any specific outbreak.

16 Real-World Interpretation

The SEIHRV framework is particularly well-suited to pathogens for which several classical SIR/SEIR assumptions break down simultaneously: SARS-CoV-2, seasonal and pandemic influenza, and cholera all involve nontrivial latency, presymptomatic spread, hospitalization dynamics, and (for SARS-CoV-2 and cholera in particular) substantial environmental transmission. We now discuss how each feature of the model maps onto observed epidemiological phenomena, and what insights the mathematical analysis provides.

Environmental persistence. SARS-CoV-2 has been detected on surfaces and in respiratory aerosols for hours to days under typical indoor conditions [22]. This empirical observation justifies the inclusion of the environmental compartment $V(t)$ in the model. The decay rate η depends on temperature, humidity, and UV exposure; in our framework, this directly induces a seasonal modulation of the effective transmission rate β_3 and thus of the $\mathcal{R}_V$ component of $\mathcal{R}_0$. For cholera, where the pathogen V corresponds literally to *Vibrio*

Table 4: Structural and dynamical comparison of the SIR and SEIHRV models.

Feature	SIR	SEIHRV
Dimension	3	6
Transmission pathways	1	3 ($\beta_1, \beta_2, \beta_3$)
Latent period	No	Yes ($1/\sigma$)
Hospitalization	No	Yes
Environmental reservoir	No	Yes
$\mathcal{R}_0$ formula	$\beta S_0/\gamma$	Eq. (10)
DFE globally stable ($\mathcal{R}_0 < 1$)	Yes (trivially)	Yes (Theorem 8.1)
Periodic orbits possible	No (Bendixson)	Yes (Proposition 10.2)
Hopf bifurcation	Impossible	Possible
Backward bifurcation	No	Possible (if $a > 0$)
Parameter uncertainty effect	$\sim$ linear in β	Nonlinear, high-dimensional

cholerae in water supplies, environmental dynamics are dominant and have been modeled extensively along the same lines [16].

Presymptomatic transmission. Contact tracing during the COVID-19 pandemic established that a substantial fraction of transmission events occurred before symptom onset [23]. In the SEIHRV model, this corresponds to the $\beta_2 SE$ term and the $\mathcal{R}_E$ component of $\mathcal{R}_0$. Such pathogens would naturally have an $\mathcal{R}_0$ that is underestimated with a standard SEIR model where $\beta_2 = 0$, which would result in under-cautious public health responses. This contribution is made explicit and quantifiable in the decomposition given in Section 6, $\mathcal{R}_0 = \mathcal{R}_E + \mathcal{R}_I + \mathcal{R}_V$.

The atmosphere and capability of the hospital. The rate of hospitalization δ and the recovery rate of the person in hospital ρ are model parameters which can be considered constant or dependent on the number of hospitalized individuals H . If the capacity of the hospital is exceeded, then ρ will decrease (stays per patient will increase), and mortality due to the disease increases, which will have a nonlinear effect on the dynamics. In the present analysis, it is assumed that ρ is a constant, but the framework can easily be generalized to a capacity-dependent recovery $\rho = \rho(H)$ that adds further nonlinearities and can result in backward bifurcations, and even multistability [4].

Seasonality, delayed waves. In the environmental feedback loop $I \rightarrow V \rightarrow E \rightarrow I$, there is a distributed delay between exposure and reinfection of the susceptible population. This is the mechanistic basis for the Hopf bifurcation discussed in Section 10 and thus, of the multi-wave structure in the COVID-19 and influenza pandemics. Importantly, the model can generate these waves automatically, without explicit time-delay differential equations or forcing terms through the parameters, being explicit seasons. Such external modulation (e.g.

β_1 or β_3 time-dependent sinusoidally) would amplify and entrain these intrinsic oscillations, which have also been found experimentally in respiratory pathogens [17].

Policy implications. The elasticity table (Section 11) gives a quantitative ranking of interventions. If $\mathcal{R}_V$ is the dominant mode (as can be assumed when the water is contaminated, e.g., by cholera), then the infrastructure for sanitation or water-treatment (i.e., reducing β_3 or increasing η) will provide the greatest reduction in $\mathcal{R}_0$ in the margin. Early testing, mask mandates, and contact tracing of presymptomatic contacts offer disproportionate benefit if $\mathcal{R}_E$ is dominant (as it seems to have been with early COVID-19). Development of the mathematical model becomes, therefore, not only a descriptive but also a prescriptive tool for making public health policy.

17 Limitations

The SEIHRV model significantly expands the classical SIR model, but certain modeling assumptions limit the applicability of the model. We explicitly spell them out here, not only to make it clear what we have studied, but also what further lines of investigation we have uncovered.

- (1) **Homogeneous mixing.** The mass-action transmission terms ($\beta_i SI$ etc.) are an assumption that each susceptible person comes into contact with each infectious person at the same per-capita rate. Real populations are socially and spatially heterogeneous [19, 10], and this can significantly influence the value of $\mathcal{R}_0$ and the time of the epidemic peak.
- (2) **Deterministic ODE assumption.** The model assumes that there is no intrinsic stochastic noise and all populations are continuous and deterministic. Small infectious populations (e.g., at the start of an epidemic or when a population of infectious individuals begins to die out) can be dominated by stochastic effects [17].
- (3) **No spatial structure.** Spatial compartments are not explicitly modeled; there is no spatial delay and transmission is global. Spatially explicit models (reaction–diffusion PDEs or patch models) have wave propagation phenomena that we do not have in the present model.
- (4) **Constant parameters in time.** All parameters $\beta_i, \sigma, \gamma, \dots$ are treated as time-invariant. In reality, transmission rates vary seasonally, behavioral responses adjust during outbreaks, and interventions change over time. The probabilistic framework of Section 12 addresses parameter *uncertainty* but not parameter *time-variation*.
- (5) **No strain evolution or waning immunity.** The recovered class R is absorbing: individuals never lose immunity, and pathogen mutation is not represented. For pathogens with multiple strains or waning immunity (such as SARS-CoV-2 with its successive variants), an SIRS-type recycling term or multi-strain extension would be required.

- (6) **Conjecture 14.1 unproven.** The numerical sweep in Section 14 provides strong evidence that the endemic equilibrium $\mathcal{E}^*$ is globally asymptotically stable whenever $\mathcal{R}_0 > 1$, but a rigorous proof via the compound-matrix approach of Li and Muldowney [15, 28] remains an open problem. Such a proof would close the gap between the algebraic possibility of Hopf bifurcation (Section 10) and the empirical absence of it (Figure 5).
- (7) **Environmental compartment is single-pool.** The viral load V is treated as a single, well-mixed reservoir. In practice, environmental contamination has spatial, temporal, and surface-dependent heterogeneity [22] that a single ODE compartment cannot capture.

These limitations are standard caveats for compartmental epidemic models and do not invalidate the qualitative conclusions of this paper. They do, however, indicate that the SEIHRV framework is a starting point rather than an endpoint: each limitation above corresponds to a recognized extension in the mathematical epidemiology literature, and combinations of these extensions remain active areas of research.

18 Conclusion

This paper has conducted a rigorous mathematical analysis of the SEIHRV epidemic model and demonstrated how increasing structural realism fundamentally transforms dynamical behavior.

Beginning from the classical SIR model, we showed that its simplicity—single transmission pathway, no latent period, no environmental reservoir—precludes oscillatory behavior and renders equilibria globally stable in a straightforward sense. The SEIHRV extension introduces a six-dimensional nonlinear dynamical system with qualitatively richer dynamics.

Our main results are:

- **Biological feasibility.** The solutions of system (3) stay nonnegative, and bounded in the compact invariant region Ω (Theorem 4.2), so system (3) is well-posed.
- **Reproduction number.** The basic reproduction number (10) was calculated using the next-generation matrix, and it is broken down into three ecologically interpretable components: direct, presymptomatic, and environmental transmission.
- **Global stability of DFE.** The Lyapunov function was defined with the NGM structure and the disease-free equilibrium is shown to be globally asymptotically stable when $\mathcal{R}_0 < 1$ (Theorem 8.1).
- **Transcritical bifurcation.** Center-manifold analysis showed that a forward transcritical bifurcation takes place at $\mathcal{R}_0 = 1$, and a unique stable endemic equilibrium state arises for $\mathcal{R}_0 > 1$.
- **Hopf bifurcation and limit cycles.** The presence of the environmental feedback loop allows the endemic equilibrium to lose stability by Hopf bifurcation, leading to

the occurrence of sustained oscillations, which cannot occur by Hopf bifurcation in the SIR model (Proposition 10.2).

- **Probabilistic outcomes.** By making the parameters random variables, a distribution is induced over the epidemic trajectories, which can result in qualitatively different trajectories (extinction, endemic persistence, or oscillation), and sensitivity indices are related to the variance in outcomes through the delta method.

The general result is that epidemic modeling isn't just about the addition of compartments for biological completeness. For every additional dimension added to the model, new bifurcations, new attractors, and new sensitivity pathways can arise, with a major impact on model predictions and intervention design. These are not just the niceties that can be added on later; they are indispensable tools for dealing with the dynamics of epidemics in high dimensions. The change from low-dimensional epidemic models to high-dimensional nonlinear systems comes not only in terms of the number of variables but also qualitatively: new compartments lead to new attractors, new bifurcations, and new definitions of predictability, etc.

Appendix

Technical computations that are referenced in the main text are detailed and verified only in the appendices below. Appendix A records the explicit inverse of the transition matrix V used in the next-generation matrix computation; Appendix B states the Routh–Hurwitz stability conditions that govern the Hopf bifurcation; Appendix C provides the term-by-term verification of the Lyapunov inequality $\dot{L} \leq (\mathcal{R}_0 - 1)k_E E$; and Appendix D attributes the figures.

A Proof Details: Computation of V^{-1}

Proposition A.1 (Explicit inverse of the transition matrix). *Let V be the transition matrix defined in (7). Then V is invertible, and its inverse is given by*

$$V^{-1} = \frac{1}{k_E k_I \eta} \begin{pmatrix} k_I \eta & 0 & 0 \\ \sigma \eta & k_E \eta & 0 \\ \sigma \alpha & k_E \alpha & k_E k_I \end{pmatrix}.$$

Proof. Since V is lower-triangular with strictly positive diagonal entries $k_E, k_I, \eta > 0$, it is invertible [13, Cor. 1.3.3]. Performing block-row elimination (or, equivalently, forward substitution column by column) on the augmented matrix $[V \mid I_3]$ yields the formula above. One verifies directly that $VV^{-1} = I_3$ by matrix multiplication. $\square$

Remark A.2 (Probabilistic interpretation). The entry $(V^{-1})_{ij}$ has a natural meaning in the next-generation formalism [2]: it represents the expected time a single individual initially in infected compartment j spends in compartment i before leaving the infected subsystem. For instance, $(V^{-1})_{31} = \sigma \alpha / (k_E k_I \eta)$ is the expected pathogen-load contribution to V generated by a single individual initially in E .

B Routh–Hurwitz Conditions for the Jacobian at $\mathcal{E}^*$

The characteristic polynomial of the sub-Jacobian $J_{\text{inf}}(\mathcal{E}^*)$ restricted to (E, I, V) is a cubic

$$p(\lambda) = \lambda^3 + a_1 \lambda^2 + a_2 \lambda + a_3,$$

with coefficients depending on the endemic equilibrium values (S^*, E^*, I^*, V^*) . By the Routh–Hurwitz criterion [9, §15.3], all roots of p have negative real part if and only if

$$a_1 > 0, \quad a_3 > 0, \quad a_1 a_2 > a_3.$$

When the shedding rate α increases past the critical value α^* , the third condition $a_1 a_2 > a_3$ is violated, a pair of complex roots crosses the imaginary axis, and a Hopf bifurcation occurs (Proposition 10.2). The transversality condition $d(\text{Re } \lambda)/d\alpha|_{\alpha^*} \neq 0$ is verified by implicit differentiation of $p(\lambda(\alpha)) = 0$ with respect to α and using the non-degeneracy of the Jacobian at the bifurcation point [12].

C Explicit Lyapunov Coefficient Computation

We verify that the Lyapunov function (14) satisfies $\dot{L} \leq 0$ by explicit calculation. Define $c_1 = 1/k_E$, $c_2 = \beta_1 S_0/(k_E k_I)$, $c_3 = \beta_3 \sigma \alpha S_0/(k_E k_I \eta)$. Then

$$\begin{aligned}\dot{L} &= c_1 \dot{E} + c_2 \dot{I} + c_3 \dot{V} \\ &= c_1[(\beta_1 I + \beta_2 E + \beta_3 V)S - k_E E] + c_2[\sigma E - k_I I] + c_3[\alpha I - \eta V].\end{aligned}$$

Using $S \leq S_0$:

$$\dot{L} \leq c_1[\beta_1 S_0 I + \beta_2 S_0 E + \beta_3 S_0 V - k_E E] + c_2 \sigma E - c_2 k_I I + c_3 \alpha I - c_3 \eta V.$$

Collecting terms in E , I , V separately and cancelling (verified term by term):

$$\dot{L} \leq (\mathcal{R}_0 - 1)k_E E \leq 0 \quad \text{when } \mathcal{R}_0 \leq 1.$$

D Figure Attribution

Figure 1 (Compartment Transfer Diagram). The transfer-diagram layout—boxes for epidemiological compartments connected by labeled arrows, with a separate environmental node linked by dashed arrows—follows the standard schematic convention used throughout the compartmental modeling literature [10, 11, 3]. The specific topology (six compartments S, E, I, H, R, V with environmental feedback) is original to this paper and was drawn in TikZ [18]. No external figure was copied or adapted.

Figures 2, 3, and 4 (numerical results). All three are original numerical results produced for this paper. Trajectories were computed by direct integration of system (3) using the LSODA solver [21] as implemented in SciPy [26]; array operations and finite-difference Jacobians for the eigenvalue analysis in Figure 4 used NumPy [27]. The figures themselves are rendered in TikZ/PGFPlots [18] from coordinate data emitted by the simulation script. Qualitative comparison with related SEIR-type models with environmental transmission can be found in Tuite et al. [16] and in the global-stability analysis of Li and Muldowney [15]; no figure from those works was reproduced. Complete reproduction code is provided as supplementary material [34].

References

- [1] Kermack, W. O. and McKendrick, A. G. (1927). A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society A*, 115(772):700–721.
- [2] Diekmann, O., Heesterbeek, J. A. P., and Metz, J. A. J. (1990). On the definition and the computation of the basic reproduction ratio R_0 in models for infectious diseases in heterogeneous populations. *Journal of Mathematical Biology*, 28(4):365–382.

- [3] van den Driessche, P. and Watmough, J. (2002). Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission. *Mathematical Biosciences*, 180(1–2):29–48.
- [4] Castillo-Chavez, C. and Song, B. (2004). Dynamical models of tuberculosis and their applications. *Mathematical Biosciences and Engineering*, 1(2):361–404.
- [5] LaSalle, J. P. (1976). *The Stability of Dynamical Systems*. SIAM, Philadelphia.
- [6] Chitnis, N., Hyman, J. M., and Cushing, J. M. (2008). Determining important parameters in the spread of malaria through the sensitivity analysis of a mathematical model. *Bulletin of Mathematical Biology*, 70(5):1272–1296.
- [7] Perko, L. (2001). *Differential Equations and Dynamical Systems*, 3rd ed. Springer, New York.
- [8] Strogatz, S. H. (2018). *Nonlinear Dynamics and Chaos*, 2nd ed. CRC Press, Boca Raton.
- [9] Murray, J. D. (2002). *Mathematical Biology I: An Introduction*, 3rd ed. Springer, New York.
- [10] Brauer, F. and Castillo-Chavez, C. (2012). *Mathematical Models in Population Biology and Epidemiology*, 2nd ed. Springer, New York.
- [11] Hethcote, H. W. (2000). The mathematics of infectious diseases. *SIAM Review*, 42(4):599–653.
- [12] Kuznetsov, Y. A. (2004). *Elements of Applied Bifurcation Theory*, 3rd ed. Springer, New York.
- [13] Horn, R. A. and Johnson, C. R. (2013). *Matrix Analysis*, 2nd ed. Cambridge University Press, Cambridge.
- [14] Grönwall, T. H. (1919). Note on the derivatives with respect to a parameter of the solutions of a system of differential equations. *Annals of Mathematics*, 20(4):292–296.
- [15] Li, M. Y. and Muldowney, J. S. (1995). Global stability for the SEIR model in epidemiology. *Mathematical Biosciences*, 125(2):155–164.
- [16] Tuite, A. R., Tien, J., Eisenberg, M., Earn, D. J. D., Ma, J., and Fisman, D. N. (2010). Cholera epidemic in Haiti, 2010: Using a transmission model to explain spatial spread of disease and identify optimal control interventions. *Annals of Internal Medicine*, 154(9):593–601.
- [17] Wearing, H. J., Rohani, P., and Keeling, M. J. (2005). Appropriate models for the management of infectious diseases. *PLOS Medicine*, 2(7):e174.
- [18] Tantau, T. (2023). *The TikZ and PGF Packages*, version 3.1.10. <https://ctan.org/pkg/pgf>.

- [19] Anderson, R. M. and May, R. M. (1991). *Infectious Diseases of Humans: Dynamics and Control*. Oxford University Press, Oxford.
- [20] Casella, G. and Berger, R. L. (2002). *Statistical Inference*, 2nd ed. Duxbury, Pacific Grove, CA.
- [21] Hairer, E., Nørsett, S. P., and Wanner, G. (1993). *Solving Ordinary Differential Equations I: Nonstiff Problems*, 2nd ed. Springer, Berlin.
- [22] van Doremalen, N., Bushmaker, T., Morris, D. H., Holbrook, M. G., et al. (2020). Aerosol and surface stability of SARS-CoV-2 as compared with SARS-CoV-1. *New England Journal of Medicine*, 382(16):1564–1567.
- [23] He, X., Lau, E. H. Y., Wu, P., et al. (2020). Temporal dynamics in viral shedding and transmissibility of COVID-19. *Nature Medicine*, 26(5):672–675.
- [24] Peng, S. (2019). *Nonlinear Expectations and Stochastic Calculus under Uncertainty*. Springer, Berlin.
- [25] Le Maître, O. P. and Knio, O. M. (2010). *Spectral Methods for Uncertainty Quantification: With Applications to Computational Fluid Dynamics*. Springer, Dordrecht.
- [26] Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272.
- [27] Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array programming with NumPy. *Nature*, 585(7825):357–362.
- [28] Li, M. Y. and Muldowney, J. S. (1996). A geometric approach to global-stability problems. *SIAM Journal on Mathematical Analysis*, 27(4):1070–1083.
- [29] Martin, R. H. (1974). Logarithmic norms and projections applied to linear differential systems. *Journal of Mathematical Analysis and Applications*, 45(2):432–454.
- [30] Butler, G., Freedman, H. I., and Waltman, P. (1986). Uniformly persistent systems. *Proceedings of the American Mathematical Society*, 96(3):425–430.
- [31] Smith, H. L. (1995). *Monotone Dynamical Systems: An Introduction to the Theory of Competitive and Cooperative Systems*. American Mathematical Society, Providence.
- [32] Muldowney, J. S. (1990). Compound matrices and ordinary differential equations. *Rocky Mountain Journal of Mathematics*, 20(4):857–872.
- [33] Li, M. Y. and Wang, L. (2002). Global stability in some SEIR epidemic models. In *Mathematical Approaches for Emerging and Reemerging Infectious Diseases: Models, Methods, and Theory*, IMA Volumes in Mathematics and its Applications, vol. 126, pp. 295–311. Springer, New York.
- [34] Upreti, P. (2026). SEIHRV simulation code (`seihrv_simulations.py`). Supplementary material to this paper. <https://github.com/pralfredo/seihrv>